

Reasoning

PGS.TS. Nguyễn Phương Thái

TS. Hoàng Thanh Tùng

TS. Trần Hồng Việt

NLP Laboratory, Institute of Artificial Intelligence

Adapted from slides of CS224N: Natural Language Processing with Deep Learning, Stanford

What is Reasoning?

Using *facts* and *logic* to arrive at an answer

What is Reasoning?

Using *facts* and *logic* to arrive at an answer

Deductive Reasoning: Use logic to go from premise to firm conclusion

Premise: All mammals have kidneys
Premise: All whales are mammals
Conclusion: All whales have kidneys

What is Reasoning?

Using *facts* and *logic* to arrive at an answer

Deductive Reasoning: Use logic to go from premise to firm conclusion

Premise: All mammals have kidneys
Premise: All whales are mammals
Conclusion: All whales have kidneys

Inductive Reasoning: From observation, predict a likely conclusion

Observation: When we see a creature with wings, it is usually a bird
Observation: We see a creature with wings.
Conclusion: The creature is likely to be a bird

What is Reasoning?

Using *facts* and *logic* to arrive at an answer

Deductive Reasoning: Use logic to go from premise to firm conclusion

Premise: All mammals have kidneys
Premise: All whales are mammals
Conclusion: All whales have kidneys

Inductive Reasoning: From observation, predict a likely conclusion

Observation: When we see a creature with wings, it is usually a bird
Observation: We see a creature with wings.
Conclusion: The creature is likely to be a bird

Abductive Reasoning: From observation, predict the most likely explanation

Observation: The car cannot start and there is a puddle of liquid under the engine.
Likely Explanation: The car has a leak in the radiator

Reasoning: Formal vs Informal

Formal Reasoning: Follows formal rules of logic along with axiomatic knowledge to derive conclusions.

Informal Reasoning: Uses intuition, experience, common sense to arrive at answers.

For most of this lecture, by “reasoning” we mean informal deductive reasoning, often involving multiple steps

Reasoning in Large Language Models

Large Language models are **REALLY GOOD** at predicting **plausible continuations of text (Lecture-9)**, that respect **constraints in the input (Lecture 10,11)**, and align **well with human preferences (Lecture-10, 11)**.

Question: Can **current** LLMs reason?

Reasoning in Large Language Models: prompting

Chain-of-thought prompting:

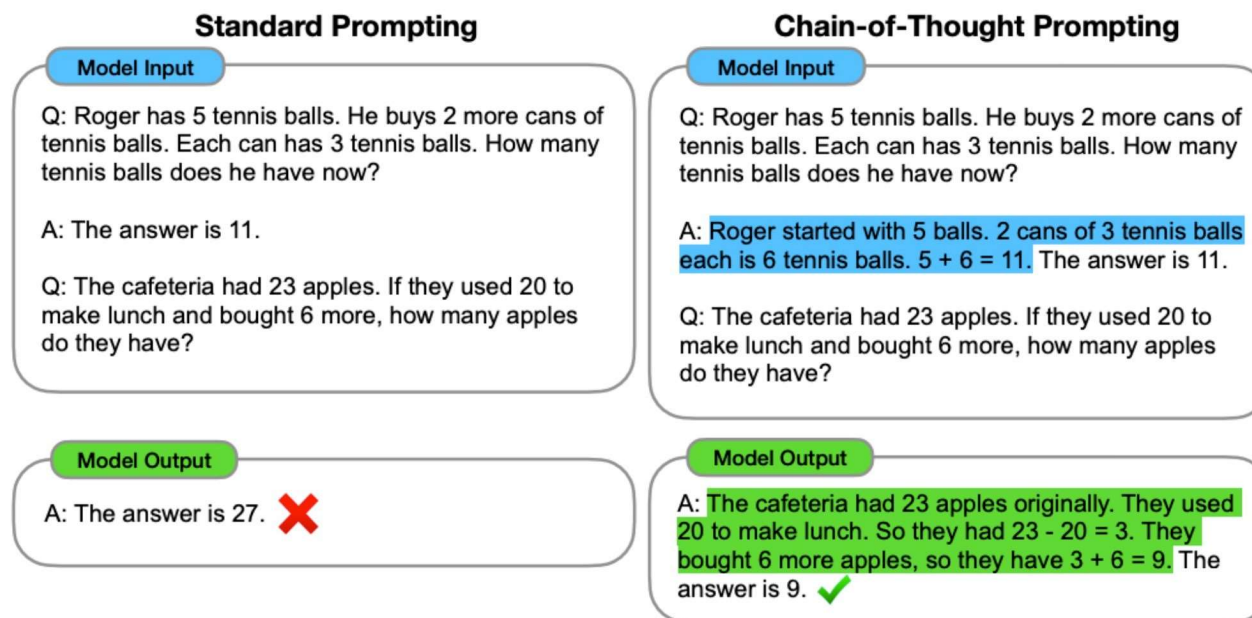


Figure 1: Chain-of-thought prompting enables large language models to tackle complex arithmetic, commonsense, and symbolic reasoning tasks. Chain-of-thought reasoning processes are highlighted.

Reasoning in Large Language Models: prompting

Zero-shot CoT prompting:

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The answer is 8. ✗

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are $16 / 2 = 8$ golf balls. Half of the golf balls are blue. So there are $8 / 2 = 4$ blue golf balls. The answer is 4. ✓

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: The answer (arabic numerals) is

(Output) 8 ✗

(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

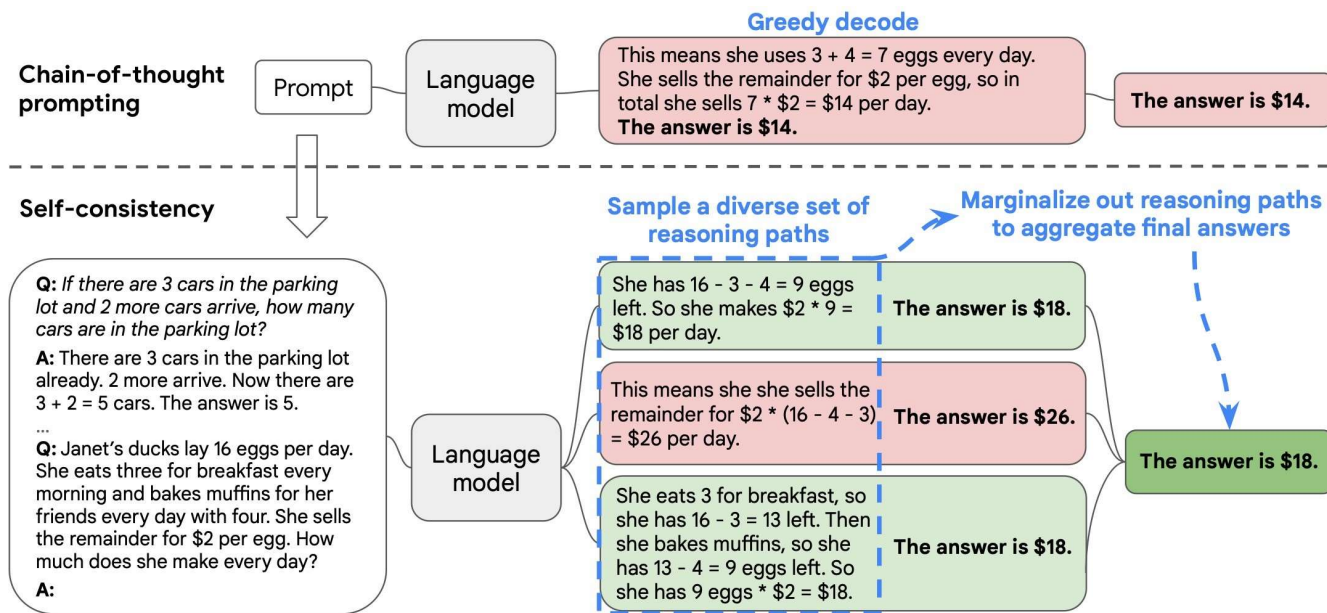
A: **Let's think step by step.**

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

Reasoning in Large Language Models: prompting

CoT with "Self-consistency": Replace greedy decoding with an ensemble of samples...

Main idea: correct reasoning processes have greater agreement than incorrect processes.



Reasoning in Large Language Models: prompting

	Method	AddSub	MultiArith	ASDiv	AQuA	SVAMP	GSM8K
	Previous SoTA	94.9^a	60.5 ^a	75.3 ^b	37.9 ^c	57.4 ^d	35 ^e / 55 ^g
UL2-20B	CoT-prompting	18.2	10.7	16.9	23.6	12.6	4.1
	Self-consistency	24.8 (+6.6)	15.0 (+4.3)	21.5 (+4.6)	26.9 (+3.3)	19.4 (+6.8)	7.3 (+3.2)
LaMDA-137B	CoT-prompting	52.9	51.8	49.0	17.7	38.9	17.1
	Self-consistency	63.5 (+10.6)	75.7 (+23.9)	58.2 (+9.2)	26.8 (+9.1)	53.3 (+14.4)	27.7 (+10.6)
PaLM-540B	CoT-prompting	91.9	94.7	74.0	35.8	79.0	56.5
	Self-consistency	93.7 (+1.8)	99.3 (+4.6)	81.9 (+7.9)	48.3 (+12.5)	86.6 (+7.6)	74.4 (+17.9)

Out-performs regular CoT on a variety of benchmarks

SELF-CONSISTENCY IMPROVES CHAIN OF THOUGHT REASONING IN LANGUAGE MODELS

Xuezhi Wang^{†‡} Jason Wei[†] Dale Schuurmans[†] Quoc Le[†] Ed H. Chi[†]
Sharan Narang[†] Aakanksha Chowdhery[†] Denny Zhou^{†§}

[†]Google Research, Brain Team

[‡]xuezhiw@google.com, [§]dennyzhou@google.com

ABSTRACT

Chain-of-thought prompting combined with pre-trained large language models has achieved encouraging results on complex reasoning tasks. In this paper, we propose a new decoding strategy, *self-consistency*, to replace the naive greedy decoding used in chain-of-thought prompting. It first samples a diverse set of reasoning paths instead of only taking the greedy one, and then selects the most consistent answer by marginalizing out the sampled reasoning paths. Self-consistency leverages the intuition that a complex reasoning problem typically admits multiple different ways of thinking leading to its unique correct answer. Our extensive empirical evaluation shows that self-consistency boosts the performance of chain-of-thought prompting with a striking margin on a range of popular arithmetic and commonsense reasoning benchmarks, including GSM8K (+17.9%), SVAMP (+11.0%), AQuA (+12.2%), StrategyQA (+6.4%) and ARC-challenge (+3.9%).

Reasoning in Large Language Models: prompting

	Method	AddSub	MultiArith	ASDiv	AQuA	SVAMP	GSM8K
	Previous SoTA	94.9^a	60.5 ^a	75.3 ^b	37.9 ^c	57.4 ^d	35 ^e / 55 ^g
UL2-20B	CoT-prompting	18.2	10.7	16.9	23.6	12.6	4.1
	Self-consistency	24.8 (+6.6)	15.0 (+4.3)	21.5 (+4.6)	26.9 (+3.3)	19.4 (+6.8)	7.3 (+3.2)
LaMDA-137B	CoT-prompting	52.9	51.8	49.0	17.7	38.9	17.1
	Self-consistency	63.5 (+10.6)	75.7 (+23.9)	58.2 (+9.2)	26.8 (+9.1)	53.3 (+14.4)	27.7 (+10.6)
PaLM-540B	CoT-prompting	91.9	94.7	74.0	35.8	79.0	56.5
	Self-consistency	93.7 (+1.8)	99.3 (+4.6)	81.9 (+7.9)	48.3 (+12.5)	86.6 (+7.6)	74.4 (+17.9)

Out-performs regular CoT on a variety of benchmarks

	GSM8K	MultiArith	SVAMP	ARC-e	ARC-c
CoT (Wei et al., 2022)	17.1	51.8	38.9	75.3	55.1
Ensemble (3 sets of prompts)	18.6 ± 0.5	57.1 ± 0.7	42.1 ± 0.6	76.6 ± 0.1	57.0 ± 0.2
Ensemble (40 prompt permutations)	19.2 ± 0.1	60.9 ± 0.2	42.7 ± 0.1	76.9 ± 0.1	57.0 ± 0.1
Self-Consistency (40 sampled paths)	27.7 ± 0.2	75.7 ± 0.3	53.3 ± 0.2	79.3 ± 0.3	59.8 ± 0.2

Self-consistency is doing more than simple ensembling

LEAST-TO-MOST PROMPTING ENABLES COMPLEX REASONING IN LARGE LANGUAGE MODELS

Denny Zhou^{†*} Nathanael Schärli[†] Le Hou[†] Jason Wei[†] Nathan Scales[†] Xuezhi Wang[†]
Dale Schuurmans[†] Claire Cui[†] Olivier Bousquet[†] Quoc Le[†] Ed Chi[†]

[†]Google Research, Brain Team

ABSTRACT

Chain-of-thought prompting has demonstrated remarkable performance on various natural language reasoning tasks. However, it tends to perform poorly on tasks which requires solving problems harder than the exemplars shown in the prompts. To overcome this challenge of easy-to-hard generalization, we propose a novel prompting strategy, *least-to-most prompting*. The key idea in this strategy is to break down a complex problem into a series of simpler subproblems and then solve them in sequence. Solving each subproblem is facilitated by the answers to previously solved subproblems. Our experimental results on tasks related to symbolic manipulation, compositional generalization, and math reasoning reveal that least-to-most prompting is capable of generalizing to more difficult problems than those seen in the prompts. A notable finding is that when the GPT-3 `code-davinci-002` model is used with least-to-most prompting, it can solve the compositional generalization benchmark SCAN in any split (including length split) with an accuracy of at least 99% using just 14 exemplars, compared to only 16% accuracy with chain-of-thought prompting. This is particularly noteworthy because neural-symbolic models in the literature that specialize in solving SCAN are trained on the entire training set containing over 15,000 examples. We have included prompts for all the tasks in the Appendix.

Reasoning in Large Language Models: prompting

Problem decomposition with Least-to-Most prompting

Stage 1: Decompose Question into Subquestions

Q: It takes Amy 4 minutes to climb to the top of a slide. It takes her 1 minute to slide down. The water slide closes in 15 minutes. How many times can she slide before it closes?

Language Model

A: To solve "How many times can she slide before it closes?", we need to first solve: "How long does each trip take?"

Reasoning in Large Language Models: prompting

Problem decomposition with Least-to-Most prompting

Stage 1: Decompose Question into Subquestions

Q: It takes Amy 4 minutes to climb to the top of a slide. It takes her 1 minute to slide down. The water slide closes in 15 minutes. How many times can she slide before it closes?

Language Model

A: To solve "How many times can she slide before it closes?", we need to first solve: "How long does each trip take?"

Stage 2: Sequentially Solve Subquestions

Subquestion 1

It takes Amy 4 minutes to climb to the top of a slide. It takes her 1 minute to slide down. The slide closes in 15 minutes.

Q: How long does each trip take?

Language Model

A: It takes Amy 4 minutes to climb and 1 minute to slide down. $4 + 1 = 5$. So each trip takes 5 minutes.

Append model answer to Subquestion 1

It takes Amy 4 minutes to climb to the top of a slide. It takes her 1 minute to slide down. The slide closes in 15 minutes.

Q: How long does each trip take?

A: It takes Amy 4 minutes to climb and 1 minute to slide down. $4 + 1 = 5$. So each trip takes 5 minutes.

Subquestion 2

Q: How many times can she slide before it closes?

Language Model

A: The water slide closes in 15 minutes. Each trip takes 5 minutes. So Amy can slide $15 \div 5 = 3$ times before it closes.

Reasoning in Large Language Models: prompting

Least-to-Most prompting for Math reasoning

Q: Elsa has 5 apples. Anna has 2 more apples than Elsa. How many apples do they have together?
A: Anna has 2 more apples than Elsa, so Anna has $2 + 5 = 7$ apples. Elsa and Anna have $5 + 7 = 12$ apples together. The answer is 12.

Standard CoT prompting

Q: Elsa has 5 apples. Anna has 2 more apples than Elsa. How many apples do they have together?
A: Let's break down this problem: 1. How many apples does Anna have? 2. How many apples do Elsa and Anna have together?
1. Anna has 2 more apples than Elsa. So Anna has $2 + 5 = 7$ apples.
2. Elsa and Anna have $5 + 7 = 12$ apples together.

Least-to-most prompting

Q: {question}
A: Let's break down this problem:

The answer is:

Reasoning in Large Language Models: prompting

Least-to-Most prompting for Math reasoning

Accuracy by Steps (GSM8K)	All	2 Steps	3 Steps	4 steps	≥ 5 steps
Least-to-Most	62.39	74.53	68.91	59.73	45.23
Chain-of-Thought	60.87	76.68	67.29	59.39	39.07

Generalizes to more #steps
than in-context example!

Reasoning in Large Language Models: prompting

Least-to-Most prompting for Math reasoning

Accuracy by Steps (GSM8K)	All	2 Steps	3 Steps	4 steps	≥ 5 steps
Least-to-Most	62.39	74.53	68.91	59.73	45.23
Chain-of-Thought	60.87	76.68	67.29	59.39	39.07

Generalizes to more #steps
than in-context example!

Prompting method	Accuracy
Zero-Shot	16.38
Standard prompting	17.06 ³
Chain-of-Thought (original)	61.18
Chain-of-Thought (1-shot)	60.88
Least-to-Most (1-shot)	62.39
Chain-of-Thought (best)	68.61³
Least-to-Most (best)	68.01

But with enough prompt
engineering, CoT $\approx$ Least-to-Most

Reasoning in ~~Large~~ Language Models via distillation

So far, we've only looked at prompting >100B parameter models for reasoning

Can we get reasoning-like behavior with smaller LMs by teaching them to imitate larger models?



Orca: Instruction-tuning small LMs with CoT Rationales

1. Collect a wide variety of instructions from the FLAN-v2 collection

Mixture Name	Sampling Algorithm	Original Size	Sampled
CoT	No Sampling	150K	150K
NIV2	Stratified Sampling	5M	440K
FLAN2021	Stratified Sampling	>28.9M	2.5M
T0	Stratified Sampling	85.7M	2M
Dialog	Skipped	22.5M	0

Table 3: Construction of our training data with 5 million samples.



Orca: Progressive Learning from Complex Explanation Traces of GPT-4

Subhabrata Mukherjee^{*†}, Arindam Mitra^{*}

Ganesh Jawahar, Sahaj Agarwal, Hamid Palangi, Ahmed Awadallah

Microsoft Research

Recent research has focused on enhancing the capability of smaller models through imitation learning, drawing on the outputs generated by large foundation models (LFMs). A number of issues impact the quality of these models, ranging from limited imitation signals from shallow LFM outputs; small scale homogeneous training data; and most notably a *lack of rigorous evaluation resulting in overestimating the small model's capability as they tend to learn to imitate the style, but not the reasoning process of LFMs*. To address these challenges, we develop Orca, a 13-billion parameter model that learns to imitate the reasoning process of LFMs. Orca learns from rich signals from GPT-4 including explanation traces; step-by-step thought processes; and other complex instructions, guided by teacher assistance from ChatGPT. To promote this progressive learning, we tap into large-scale and diverse imitation data with judicious sampling and selection. Orca surpasses conventional state-of-the-art instruction-tuned models such as Vicuna-13B by more than 100% in complex zero-shot reasoning benchmarks like BigBench Hard (BBH) and 42% on AGIEval. Moreover, Orca reaches parity with ChatGPT on the BBH benchmark and shows competitive performance (4 pts gap with optimized system message) in professional and academic examinations like the SAT, LSAT, GRE, and GMAT, both in zero-shot settings without CoT; while trailing behind GPT-4. Our research indicates that learning from step-by-step explanations, whether these are generated by humans or more advanced AI models, is a promising direction to improve model capabilities and skills.

Orca: Instruction-tuning small LMs with CoT Rationales

1. Collect a wide variety of instructions from the FLAN-v2 collection.
2. Prompt GPT4 or ChatGPT with these instructions along with a **system message**

System Instruction: You are an AI assistant. User will give you a task. Your goal is to complete the task as faithfully as you can. While performing the task think step-by-step and justify your steps.

User Instruction: Use the given data to calculate the median.

Input: [7, 3, 8, 2, 10]

Output: To calculate the median, I will follow these steps:

Step 1: Arrange the data in ascending order [2, 3, 7, 8, 10]

Step 2: Determine if the data set has an odd or even number of values

The data set has 5 values, which is an odd number.

Step 3: Find the middle value

Since the data set has an odd number of values, the median is the middle value in the ordered list. The middle value in the list [2, 3, 7, 8, 10] is 7.

Orca: Instruction-tuning small LMs with CoT Rationales

1. Collect a wide variety of instructions from the FLAN-v2 collection.
2. Prompt GPT4 or ChatGPT with these instructions along with a **system message**
3. Finetune Llama-13b on outputs generated via ChatGPT + GPT4

Orca: Instruction-tuning small LMs with CoT Rationales

Evaluation: *BigBench-hard*, a collection of 23 Big-Bench tasks with a special focus on multi-step reasoning

Orca: Instruction-tuning small LMs with CoT Rationales

Evaluation: **BigBench-hard**, a collection of 23 Big-Bench tasks with a special focus on multi-step reasoning

Boolean Expressions

Q: True and False and not True and True is

A: Let's think step by step.

Remember that (i) expressions inside brackets are always evaluated first and that (ii) the order of operations from highest priority to lowest priority is "not", "and", "or", respectively.

We first simplify this expression "Z" as follows: "Z = True and False and not True and True = A and B" where "A = True and False" and "B = not True and True".

Let's evaluate A: A = True and False = False.

Let's evaluate B: B = not True and True = not (True and True) = not (True) = False.

Plugging in A and B, we get: Z = A and B = False and False = False. So the answer is False.

Orca: Instruction-tuning small LMs with CoT Rationales

Evaluation: **BigBench-hard**, a collection of 23 Big-Bench tasks with a special focus on multi-step reasoning

Data Understanding

Q: Tomorrow is 11/12/2019. What is the date one year ago from today in MM/DD/YYYY?

Options:

- (A) 09/04/2018
- (B) 11/11/2018
- (C) 08/25/2018
- (D) 11/02/2018
- (E) 11/04/2018

A: Let's think step by step.

If tomorrow is 11/12/2019, then today is 11/11/2019. The date one year ago from today is 11/11/2018. So the answer is **(B)**.

Orca: Instruction-tuning small LMs with CoT Rationales

Evaluation: **BigBench-hard**, a collection of 23 Big-Bench tasks with a special focus on multi-step reasoning

Geometric Shapes

Q: This SVG path element `<path d="M 14.19,26.04 L 51.43,39.21 L 58.44,36.69 L 56.63,30.17 L 48.53,26.66 L 14.19,26.04"/>` draws a

Options:

- (A) circle
- (B) heptagon
- (C) hexagon
- (D) kite
- (E) line
- (F) octagon
- (G) pentagon
- (H) rectangle
- (I) sector
- (J) triangle

A: Let's think step by step.

This SVG path element contains "M" and "L" commands. M takes two parameters (x,y) and moves the current point to the coordinates (x,y). L takes two parameters (x,y) and draws a line from the previous coordinate to the new coordinate (x,y).

This path can be decomposed into 6 separate commands.

- (1) M 14.19,26.04: Move the current point to 14.19,26.04.
- (2) L 51.43,39.21: Create a line from 14.19,26.04 to 51.43,39.21.
- (3) L 58.44,36.69: Create a line from 51.43,39.21 to 58.44,36.69.
- (4) L 56.63,30.17: Create a line from 58.44,36.69 to 56.63,30.17.
- (5) L 48.53,26.66: Create a line from 56.63,30.17 to 48.53,26.66.
- (6) L 14.19,26.04: Create a line from 48.53,26.66 to 14.19,26.04.

This SVG path starts at point 14.19,26.04, creates five consecutive and touching lines, and then returns back its starting point, thereby creating a five-sided shape. It does not have any curves or arches. "pentagon" is the only five-sided polygon on the list. So the answer is **(G)**.

Orca: Instruction-tuning small LMs with CoT Rationales

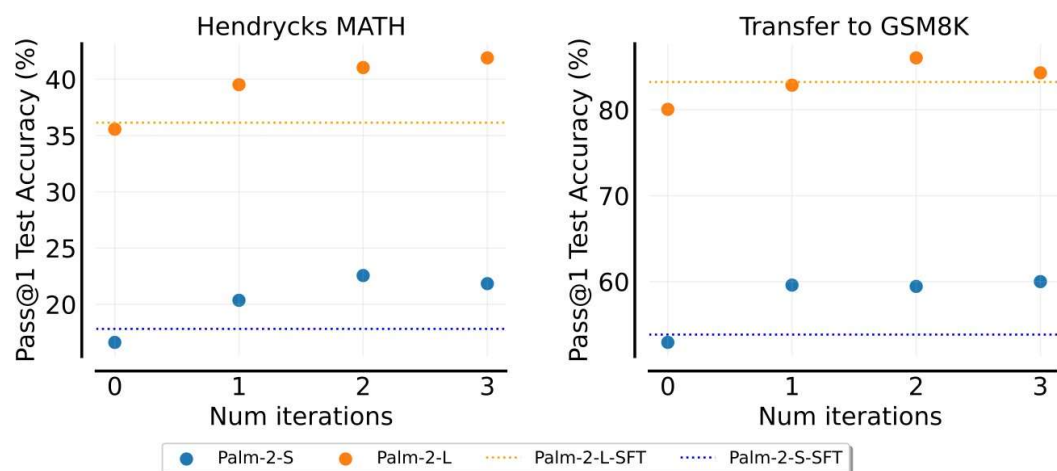
Task	ChatGPT	GPT-4	Vicuna-13B	Orca-13B
Boolean Expressions	82.8	77.6	40.8	72.0 (76.5%)
Causal Judgement	57.2	59.9	42.2	59.9 (41.8%)
Date Understanding	42.8	74.8	10.0	50.0 (400.0%)
Disambiguation QA	57.2	69.2	18.4	63.6 (245.7%)
Formal Fallacies	53.6	64.4	47.2	56.0 (18.6%)
Geometric Shapes	25.6	40.8	3.6	20.8 (477.8%)
Hyperbaton	69.2	62.8	44.0	64.0 (45.5%)
Logical Deduction (5 objects)	38.8	66.8	4.8	39.6 (725.0%)
Logical Deduction (7 objects)	39.6	66.0	1.2	36.0 (2900.0%)
Logical Deduction (3 objects)	60.4	94.0	16.8	57.6 (242.9%)
Movie Recommendation	55.4	79.5	43.4	78.3 (80.6%)
Navigate	55.6	68.8	46.4	57.6 (24.1%)
Penguins in a Table	45.9	76.7	15.1	42.5 (181.8%)
Reasoning about Colored Objects	47.6	84.8	12.0	48.4 (303.3%)
Ruin Names	56.0	89.1	15.7	39.5 (151.2%)
Salient Translation Error Detection	40.8	62.4	2.0	40.8 (1940.0%)
Snarks	59.0	87.6	28.1	62.4 (122.0%)
Sports Understanding	79.6	84.4	48.4	67.2 (38.8%)
Temporal Sequences	35.6	98.0	16.0	72.0 (350.0%)
Tracking Shuffled Objects (5 objects)	18.4	25.2	9.2	15.6 (69.6%)
Tracking Shuffled Objects (7 objects)	15.2	25.2	5.6	14.0 (150.0%)
Tracking Shuffled Objects (3 objects)	31.6	42.4	23.2	34.8 (50.0%)
Web of Lies	56.0	49.6	41.2	51.2 (24.3%)
Average	48.9	67.4	23.3	49.7 (113.7%)

- Outperforms Vicuna-13B
- Outperforms ChatGPT!
- GPT-4 has potential data contamination issues with Bigbench-hard

Reasoning by Finetuning LMs on their own outputs?

ReST^{EM} alternates between the following two steps:

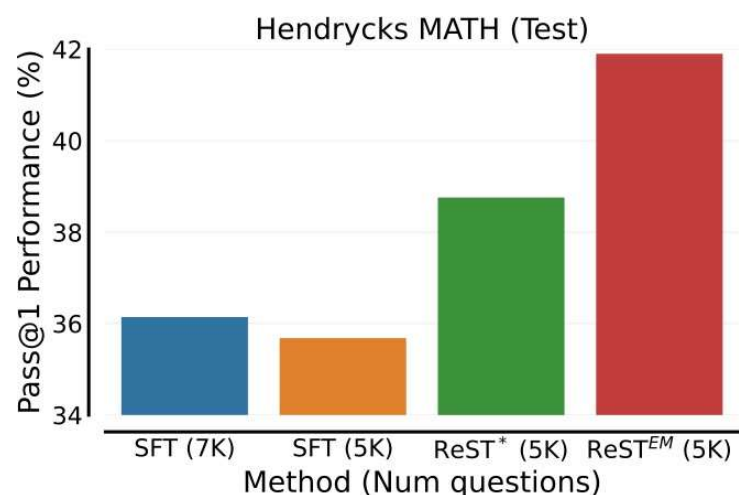
1. *Generate (E-Step)*: Given reasoning problem, sample multiple solutions from language model. Filter based on some (problem specific) function [answer correctness for math problems]
2. *Improve (M-Step)*: Update the language model to maximize probability of filtered solutions, using supervised finetuning



Reasoning by Finetuning LMs on **their own outputs**?

ReST^{EM} alternates between the following two steps:

1. *Generate (E-Step)* : Given reasoning problem, sample multiple solutions from language model. Filter based on some (problem specific) function [answer correctness for math problems]
2. *Improve (M-Step)* : Update the language model to maximize probability of filtered solutions, using supervised finetuning



Can Language Models Reason?

The Economist
Science and technology | Generative AI
Large language models' ability to generate text also lets them plan and reason

TechTalks

Large language models have a reasoning problem

?

The New York Times

Microsoft Says New A.I. Shows Signs of Human Reasoning

?

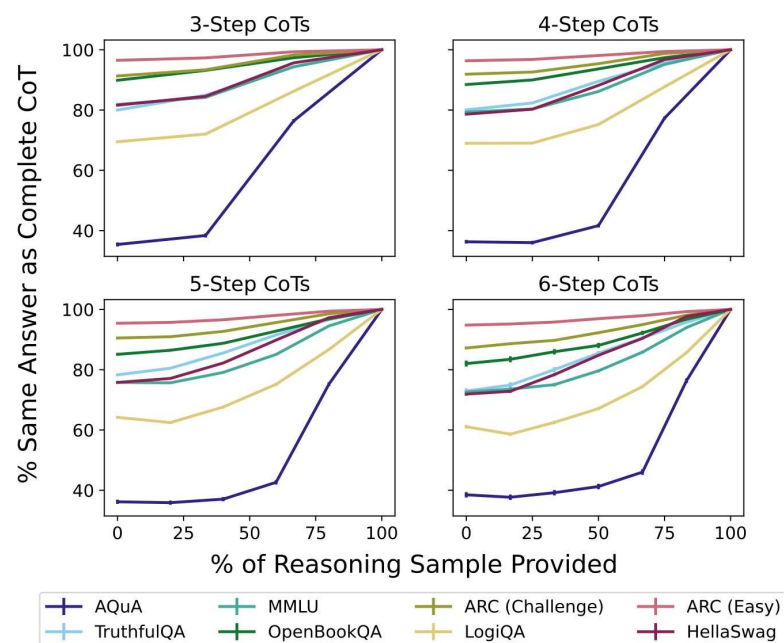
WIRED

Some Glimpse AGI in ChatGPT. Others Call It a Mirage

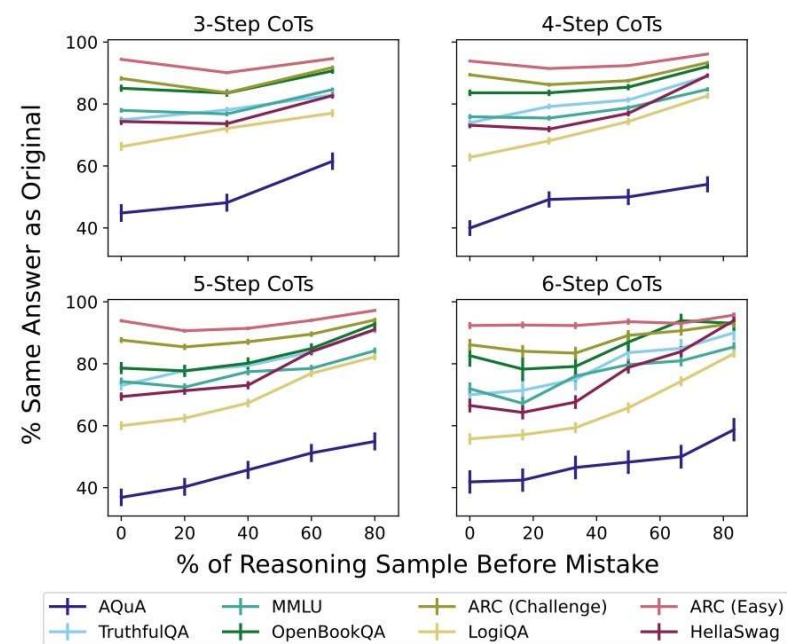
Let's look at some more careful evaluation to see if reasoning in LMs is **systematic**

Can Language Models Reason?

CoT Rationales are often not faithful



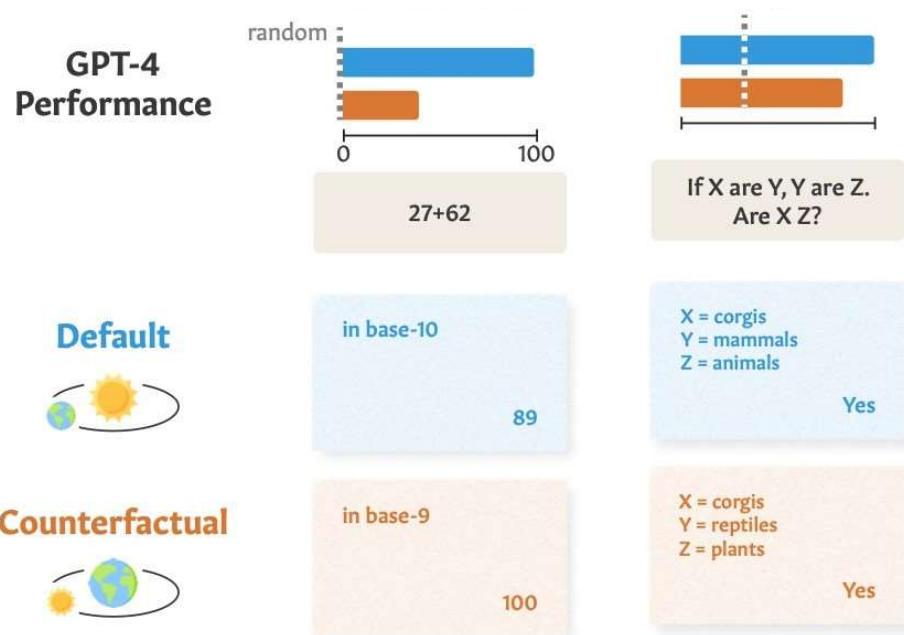
Models do not always need the full rationale to answer correctly → rationale may be post-hoc?



Sometimes, models answer correctly even with an incorrect rationale..

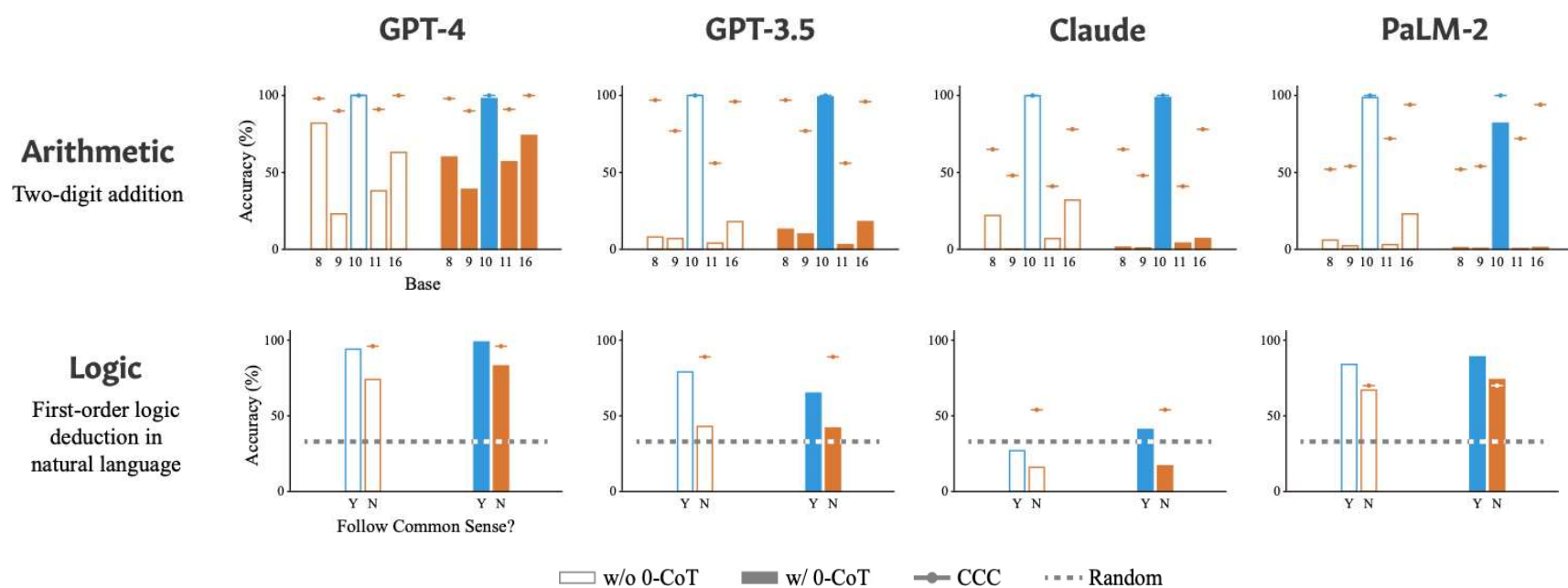
Reasoning or Reciting? Exploring the Capabilities and Limitations of Language Models Through Counterfactual Tasks

Zhaofeng Wu[Ⓜ] Linlu Qiu[Ⓜ] Alexis Ross[Ⓜ] Ekin Akyürek[Ⓜ] Boyuan Chen[Ⓜ]
Bailin Wang[Ⓜ] Najoung Kim[Ⓜ] Jacob Andreas[Ⓜ] Yoon Kim[Ⓜ]
[Ⓜ]MIT [Ⓜ]Boston University
zfw@csail.mit.edu



Can Language Models Reason?

Reasoning vs Memorization: Using Counterfactuals



Can Language Models Reason?

Reasoning vs Memorization: Counterfactuals for Analogical Reasoning

Original transformation types

Extend sequence	Successor	Predecessor
a b c d → a b c d e i j k l → i j k l m	a b c d → a b c e i j k l → i j k m	b c d e → a c d e i j k l → h j k l
Remove redundant letter	Fix alphabetic sequence	Sort
a b b c d e → a b c d e i j k k l m → i j k l m	a b c w e → a b c d e i j k x m → i j k l m	a d c b e → a b c d e k j m l i → i j k l m

Modified transformation types

Extend sequence	Successor	Predecessor
a b c d → a b c d f i j k l → i j k l n	a b c d → a b c f i j k l → i j k n	c d e f → a d e f j k l m → h k l m
Remove redundant letter	Fix alphabetic sequence	Sort
a c e g i i → a c e g i i k k m o q → i k m o q	a c e g o → a c e g i i k x o q → i k m o q	k f a p u → a f k p u i m k o q → i k m o q

Can Language Models Reason?

Reasoning vs Memorization: Counterfactuals for Analogical Reasoning

Original transformation types

Extend sequence	Successor	Predecessor
a b c d → a b c d e	a b c d → a b c e	b c d e → a c d e
i j k l → i j k l m	i j k l → i j k m	i j k l → h j k l
Remove redundant letter	Fix alphabetic sequence	Sort
a b b c d e → a b c d e	a b c w e → a b c d e	a d c b e → a b c d e
i j k k l m → i j k l m	i j k x m → i j k l m	k j m l i → i j k l m

Modified transformation types with synthetic alphabet

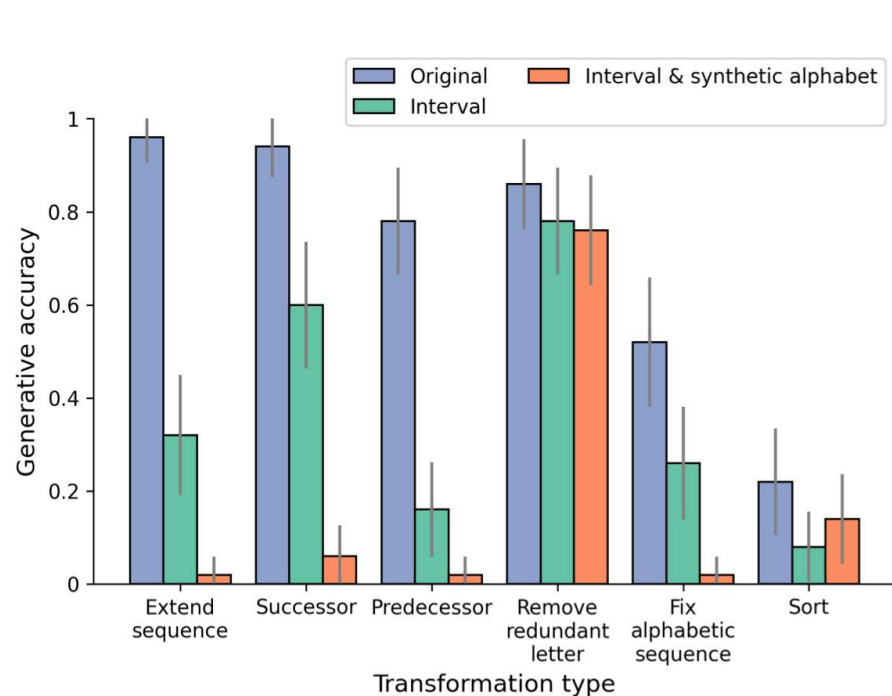
Synthetic alphabet

x y l k w b f z t n j r q a h v g m u o p d i c s e

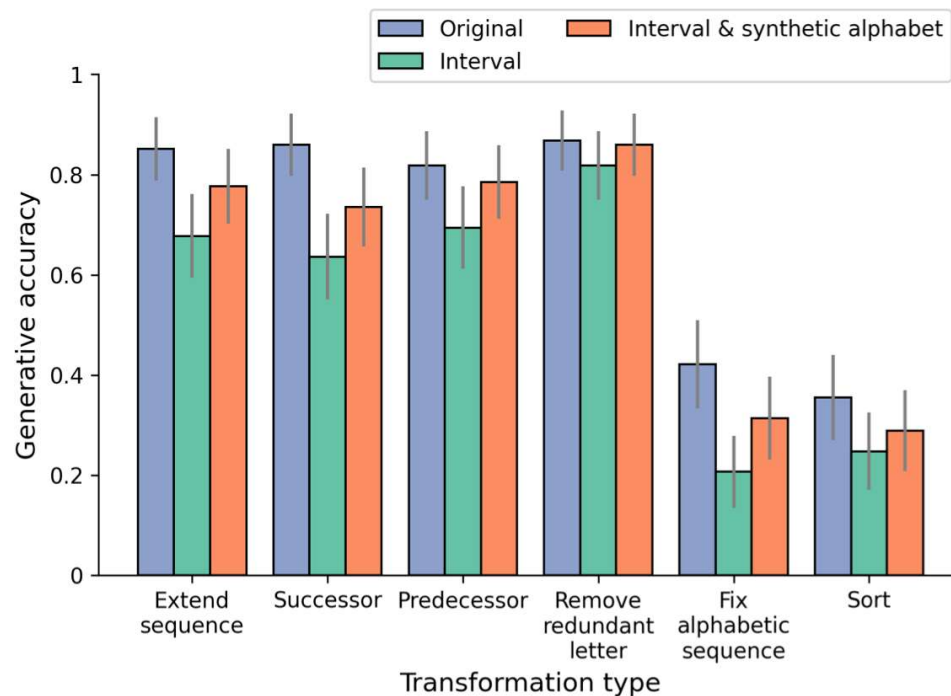
Extend sequence	Successor	Predecessor
x y l k → x y l k b	x y l k → x y l b	l k w b → x k w b
t n j r → t n j r a	t n j r → t n j a	n j r q → z j r q
Remove redundant letter	Fix alphabetic sequence	Sort
x l w w f t → x l w f t	x l w r t → x l w f t	x l f w t → x l w f t
t t j q h g → t j q h g	t j p h g → t j q h g	j t q h g → t j q h g

Can Language Models Reason?

Reasoning vs Memorization: Counterfactuals for Analogical Reasoning



Significant drop in performance for GPT-4 → evidence of spurious reasoning?



No drop in performance for humans

Lecture-13 Recap

- Reasoning in Language Models:
 - Via prompting
 - By distilling rationales from big LMs into small LMs
 - By finetuning LMs on their own rationales, iteratively
 - Counterfactual evaluation reveals reasoning may not be systematic

💡 Lots to be done to drive further improvements!